

Control Plan — Part 34521-A — Rev B (SUPERSEDED)

Rev B · effective 2024-02-15 · **SUPERSEDED by Rev C on 2025-08-01**. Retained for traceability only.
Do not use for production decisions.

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--- METADATA ---
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owner: S. Okafor, Quality Engineering — Cleveland
part_numbers: [34521-A]
operation_numbers: [OP10, OP20, OP30, OP40]
machine_ids: [BSAW-03, NLX-08, NLX-09]
controller_models: [Fanuc 0i-TF]
fault_codes: []
materials: [MAT-4140]
standards: [IATF-16949-2016 §8.5.1.1]
supplier_ids: [SUP-OHIO-STEEL]
jurisdictions: []
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Status

THIS REVISION IS SUPERSEDED. Rev C is the controlled document as of 2025-08-01. See [QMS-CP-34521A-RevC](#) . Any agent or process answering a question about Part 34521-A control limits, inspection frequencies, or reaction plans must cite Rev C, not this Rev B.

1. Differences vs. Rev C

Rev C tightens the bore-gauge inspection frequency at OP30 from *every 25 pieces* (Rev B) to *every 10 pieces*, and adds an explicit cross-reference to PFMEA failure mode FM-007. Rev C also formalises the 4-hour engineering disposition window per IATF 16949 §10.2.1 (this Rev B referenced only a generic “same shift” window).

2. Control plan matrix (as it stood at Rev B — for reference only)

Op	Process	Machine	Characteristic	Spec / tolerance	Method	Sample	Frequency
OP10	Stock saw cutoff	BSAW-Ø3	Bar length	250 mm ±0.5 mm	Caliper	3	Per lot
OP20	Rough turn OD	NLX-Ø8 / NLX-Ø9	OD	Ø 50.0 ±0.1 mm	Micrometer	5	Every 25 pcs
OP30	Finish turn + bore	NLX-Ø8 / NLX-Ø9	Bore diameter	Ø 38.000 +0.025 / -0.000 mm	Bore gauge	5	Every 25 pcs <i>(superseded — now every 10 pcs in Rev C)</i>
OP40	Final inspection	CMM	Bore Ø + concentricity	per OP30; concentricity 0.05 mm TIR	CMM program 34521A-CMM-001	100%	Every piece

Document control / sign-off

Role	Name	Signature	Date
Document Owner	S. Okafor, Quality Engineering — Cleveland	(on file)	2024-02-15
Approver	S. Okafor, Quality Engineering — Cleveland	(on file)	2024-02-15